

Reference-Price Sensitivity Study for the Hybrid LSMC–PDE Method

BGK-style calibrated gDMR model with 12 Bermudan exercise dates

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1. Setting and Scope

This note summarizes the experiments collected in the folder `Experiments 26.03` for the BGK-style calibrated gDMR model with parameters

$$\begin{aligned} S_0 &= 100, & v_0 &= 0.114, & v'_0 &= 0.110, & r &= 0, & T &= 1, \\ \kappa_1 &= 5.5, & \kappa_2 &= 0.1, & \theta &= 0.078, & \xi_1 &= 2.689, & \xi_2 &= 0.502, \end{aligned}$$

and

$$\delta_1 = \delta_2 = 0.94, \quad \rho_{12} = -0.982, \quad \rho_{13} = -0.727, \quad \rho_{23} = 0.59.$$

The Bermudan option is sampled at 12 exercise dates and the strike levels are $K \in \{90, 100, 110\}$, corresponding to OTM, ATM, and ITM puts.

The purpose of the note is to compare two benchmark-reference regimes:

- a reference generated with 600 Euler steps, which was used during the broader exploratory phase;
- a stronger reference generated with 1200 Euler steps and 1,200,000 benchmark paths.

The main emphasis is on the *direct estimator*, because this is where the Hybrid LSMC–PDE method displayed its clearest advantage over plain LSMC.

2. Benchmark References at 600 and 1200 Steps

Table 1 compares the benchmark direct reference prices under the two reference regimes.

Table 1: Benchmark direct reference values under the two benchmark regimes.

Scenario	600-step reference	SE	1200-step reference	SE
ATM	11.371204	0.016590	11.356823	0.015058
ITM put	16.637388	0.018957	16.645784	0.017243
OTM put	7.426156	0.013741	7.402999	0.012459

The shift from 600 to 1200 Euler steps changes the benchmark reference values only modestly, but it is still useful to check whether the ranking between LSMC and the hybrid method is stable under this stronger benchmark.

3. Experiments Using the 600-Step Reference

The 600-step reference was used for the broader experimental campaign, including the original three-scenario comparison, the hybrid tuning study, and the 48-step path-count sweeps for ATM and OTM.

3.1 Original and Tuned Three-Scenario Comparisons

Table 2: Direct prices and direct relative errors under the 600-step reference.

Scenario	Benchmark direct	Baseline hybrid	Baseline err.	Tuned hybrid	Tuned err.
ATM	11.371204	11.245782	1.103%	11.359836	0.100%
ITM put	16.637388	16.571453	0.396%	16.680641	0.260%
OTM put	7.426156	7.295876	1.754%	7.414778	0.153%

The untuned 12-date hybrid setup was already reasonably close in the ITM case, but it was above 1% in ATM and OTM. After tuning, the direct relative errors fell to 0.100% for ATM and 0.153% for OTM, while ITM remained at 0.260%.

3.2 Hybrid Tuning Scorecard

Table 3: Hybrid tuning scorecard under the 600-step reference, focusing on ATM and OTM.

Setting	ATM dir. err.	OTM dir. err.	Max err.	Average err.
baseline_20k_181	1.103%	1.754%	1.754%	1.429%
paths40k_grid181	0.075%	0.318%	0.318%	0.196%
paths40k_grid241	0.058%	0.304%	0.304%	0.181%
paths60k_grid241	0.210%	0.274%	0.274%	0.242%
paths60k_grid301	0.211%	0.282%	0.282%	0.246%
paths60k_grid301_wide_q999	0.100%	0.153%	0.153%	0.127%

The best common hybrid configuration was the 60,000-path, 301-point grid with the wider asset range and volatility quantile 0.999. This setting produced the smallest worst-case direct error over the ATM and OTM focus cases.

3.3 Path-Count Study at 48 Euler Steps

With Euler steps fixed at 48, the tuned hybrid method consistently outperformed the benchmark LSMC method in direct relative error for both ATM and OTM across the full tested path range.

Table 4: Representative direct relative errors at 48 Euler steps under the 600-step reference.

Scenario	Paths	Benchmark dir. err.	Hybrid dir. err.
ATM	1,000	5.919%	1.491%
ATM	10,000	1.678%	1.086%
ATM	20,000	1.207%	0.074%
ATM	60,000	1.104%	0.176%
OTM put	1,000	7.504%	2.585%
OTM put	10,000	2.556%	1.178%
OTM put	20,000	1.855%	0.064%
OTM put	60,000	1.452%	0.213%

This was the clearest favorable result for the hybrid method: on a coarse time grid, the hybrid direct estimator dominated the plain LSMC direct estimator over the entire path sweep in both ATM and OTM.

4. Experiments Using the 1200-Step Reference

The stronger 1200-step benchmark reference was then used to reassess the direct-error story. This is a more demanding comparison because the benchmark target is refined further in time and estimated with 1,200,000 paths.

4.1 Rebased Path-Count Results at 48 Euler Steps

The key question is whether the hybrid advantage observed under the 600-step reference survives after rebasing to the stronger benchmark. The answer is yes: the hybrid method remains more accurate than plain LSMC at all representative path counts shown below.

Table 5: Representative direct relative errors at 48 Euler steps under the 1200-step reference.

Scenario	Paths	Benchmark dir. err.	Hybrid dir. err.
ATM	1,000	6.053%	1.620%
ATM	10,000	1.806%	0.960%
ATM	20,000	1.336%	0.052%
ATM	60,000	1.232%	0.302%
OTM put	1,000	7.840%	2.905%
OTM put	10,000	2.877%	0.869%
OTM put	20,000	2.173%	0.248%
OTM put	60,000	1.769%	0.526%

Relative errors do change after rebasing, as expected, but the qualitative ranking is preserved: at 48 Euler steps the hybrid direct estimator still remains below the benchmark LSMC direct estimator across the tested path counts for both ATM and OTM.

4.2 Saved Higher-Step Check at 60 Euler Steps

An additional saved comparison is available at 60 Euler steps. Rebased to the 1200-step reference, this check also remains favorable to the hybrid method.

Table 6: Direct relative error at 60 Euler steps under the 1200-step reference.

Scenario	Benchmark direct	Hybrid direct	Benchmark dir. err.	Hybrid dir. err.
ATM	11.381616	11.350391	0.218%	0.057%
OTM put	7.435732	7.410080	0.442%	0.096%

The saved 60-step comparison suggests that the hybrid advantage is not confined to the most extreme coarse-grid case. Even after strengthening the benchmark reference, the hybrid direct estimator remains more accurate than plain LSMC in both ATM and OTM for this saved higher-step check.

5. Overall Conclusions

The experiments support three main conclusions.

- The 600-step reference regime provided the broader exploratory evidence. Within that regime, the tuned Hybrid LSMC–PDE method achieved direct relative errors below 0.3% in ATM, ITM, and OTM, and showed a strong advantage over plain LSMC in the 48-step path-count study.
- Replacing the benchmark reference by a stronger 1200-step, 1.2 million-path benchmark does not overturn the main favorable direct-error conclusions. In particular, the hybrid method remains more accurate than plain LSMC in the saved ATM and OTM path-count studies at 48 steps.
- The saved 60-step check indicates that this advantage can persist when the time grid is moderately refined, at least for the ATM and OTM cases available on disk.

Taken together, the two-reference comparison indicates that the main favorable conclusion for the Hybrid LSMC–PDE method is robust: its direct estimator performs particularly well on relatively coarse Euler grids, and this remains visible even when the benchmark target is strengthened from 600 to 1200 Euler steps.